

```
Untitled-1 lab1.py X
C:\Users\sudee> lab1.py > ...
1  from collections import deque
2
3  # Function to perform BFS
4  def bfs(graph, start):
5      visited = set()
6      queue = deque([start])
7      visited.add(start)
8      traversal = []
9
10     while queue:
11         vertex = queue.popleft()
12         traversal.append(vertex)
13
14         for neighbor in graph[vertex]:
15             if neighbor not in visited:
16                 visited.add(neighbor)
17                 queue.append(neighbor)
18
19     return traversal
20
21
```

```
lab1.py X  
C:\Users\sudee> cd lab1.py > ...  
4 def bfs(graph, start):  
18     return traversal  
19  
20  
21  
22 # Driver Code  
23 graph = {  
24     'A': ['B', 'C'],  
25     'B': ['D', 'E'],  
26     'C': ['F'],  
27     'D': [],  
28     'E': ['F'],  
29     'F': []  
30 }  
31  
32 start_node = 'A'  
33  
34 result = bfs(graph, start_node)  
35  
36 print("Breadth First Search (BFS) Traversal:")  
37 print("-> ".join(result))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

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Python + - [ ] [X] [Y]  
  
PS C:\Users\sudee\AppData\Local\Programs\Microsoft VS Code> & C:\msys64\ucrt64\bin\python.exe c:/Users/sudee/lab1.py  
Breadth First Search (BFS) Traversal:  
A -> B -> C -> D -> E -> F  
PS C:\Users\sudee\AppData\Local\Programs\Microsoft VS Code>
```